ϕ-Dark Matter in SSEE-V3.6: Algebraic Mass Derivation and Two-Sector Proposal for the $f\sigma_8$ Growth Tension

SSEE Series — Paper 6

Mike Edison Almeida Vallejo

mike.almeida1721@gmail.com

ORCID: 0009-0008-2195-7836

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Abstract

The Structural Self-Energy Expansion (SSEE-V3.6) framework derives all cosmological parameters algebraically from the golden ratio φ and π, achieving a zero-free-parameter dark energy background (Papers 1–5) that has demonstrated strong coincidence with CMB, BAO, and cluster tests. (The two-sector dark matter extension proposed here introduces $m_\phi = 5.60\,\mathrm{eV}$ as a phenomenological ansatz within the dark matter sector, not as a new free parameter of the dark energy background.) Evaluated against six canonical RSD survey points analysed here, the single-sector SSEE-V3.6 model (Papers 1–5, $\Omega_{\mathrm{m,dyn}} = 0.160$) yields a mean $f\sigma_8$ tension of 2.56σ, tracing to the dynamical matter density ($\sigma_{8,\mathrm{base}} = 0.811 \times G_{\mathrm{single}} = 0.702$; Paper 5 reports 2.67σ using a partially overlapping survey selection — consistent within survey-bin choice). We systematically scan EFT extensions (kinetic braiding α_B, run-rate α_M) and find that kinetic braiding alone is a no-go: it suppresses $f\sigma_8$ further. We then propose a *two-sector* dark matter model in which $\Omega_{\mathrm{CDM}} = 0.160$ is supplemented by a ϕ-dark matter component $\Omega_{\phi\text{-DM}} = (\mathcal{M}_{\mathrm{SSEE}} - 1)\,\Omega_{\mathrm{m,dyn}} = 0.160$, active only on scales $k < k_{\mathrm{fs}}$. We define a characteristic mass scale $m_\phi = \Sigma m_\nu \times H_0^{\mathrm{alg}} = 5.60\,\mathrm{eV}$ as a phenomenological ansatz fixed by quantities derived in Paper 4. On scales $k < k_{\mathrm{fs}}$ relevant to RSD surveys and σ_8, both dark matter sectors cluster and the effective growth density is $\Omega_{\mathrm{m,growth}} = 0.320$. The resulting two-sector growth factor $G_{2\mathrm{s}} = D_1^{\mathrm{SSEE}}/D_1^{\Lambda\mathrm{CDM}} = 0.979$ gives $\sigma_8^{\mathrm{eff}} = 0.811 \times G_{2\mathrm{s}} = 0.794$. Within the restricted parameterization used here, the two-sector model reduces the mean $f\sigma_8$ tension from 2.56σ to $\mathbf{0.50}\sigma$ across six surveys without introducing an additional continuously fitted background parameter. The proposed two-sector extension is internally consistent at the level of the linear-growth analysis presented here. The

background CMB, BAO, and cluster-sector results from Papers 2–5 are not modified by the present construction. The algebraic mass $m_\phi = 5.60\,\mathrm{eV}$ fixes a free-streaming scale $k_{\rm fs} = 0.493\,h/\mathrm{Mpc}$, which is the falsifiable target: ϕ-DM is modelled as a gravitationally produced scalar with no Standard-Model portal, so it leaves an imprint on the matter power spectrum rather than a neutrino-sector signature. We explicitly note that this is a phenomenological resolution at the linear-growth level; the first-principles relic abundance and the Lyman-α constraints on $k_{\rm fs}$ remain open challenges.

Contents

1 Introduction

The Structural Self-Energy Expansion (SSEE-V3.6) framework [Almeida Vallejo, 2026a] is a zero-free-parameter dark energy model in which the equation-of-state parameters $(w_0, w_a) = (-0.840, -0.670)$ are derived algebraically from φ (golden ratio) and π. The framework has passed four independent observational tests:

1. **Paper 2** [Almeida Vallejo, 2026b]: DESI DR2 [Abdul-Karim et al., 2025] BAO (0.05σ) and galaxy cluster mass discrepancies $(\chi^2_r = 0.122)$.

2. **Paper 3** [Almeida Vallejo, 2026c]: CMB Planck PR4 (plik_lite $\Delta\mathrm{BIC} = -31.3$ in favour of SSEE-V3.6 vs. ΛCDM).

3. **Paper 4** [Almeida Vallejo, 2026d]: Algebraic derivation of n_s, H_0, Ω_m, $\Omega_b h^2$ from the Nine Sovereignties.

4. **Paper 5** [Almeida Vallejo, 2026e]: Israel-Stewart [Israel and Stewart, 1979] causal gradient stability ($c^2_{s,\mathrm{eff}} = 0$ exact), $S_8 = 0.725$ $(1.96\sigma$ KiDS vs. 3.27σ for ΛCDM).

1.1 The residual tension

The single-sector SSEE-V3.6 model of Papers 1–5 ($\Omega_{\mathrm{m,dyn}} = 0.160$) yields a mean $f\sigma_8$ tension of 2.56σ against the six RSD survey points analysed in this paper [Beutler et al., 2012, Howlett et al., 2015, Alam et al., 2017, Hou et al., 2021, Blake et al., 2012] ($\sigma_{8,\mathrm{base}} = 0.702$; Paper 5 reports 2.67σ with a partially overlapping survey selection). The root cause is the dynamical matter density $\Omega_{\mathrm{m,dyn}} = 1 + w_0 = 0.160$, which suppresses the growth rate $f(z) \approx \Omega_m(z)^\gamma$ [Linder, 2005] at all redshifts. Crucially, Paper 5 also showed that S_8 is *better* predicted by SSEE-V3.6 than by ΛCDM, creating an S_8–$f\sigma_8$ split: the framework predicts the amplitude of fluctuations correctly but underestimates the growth rate.

This paper presents a phenomenological two-sector proposal that reduces the $f\sigma_8$ tension within the zero-parameter philosophy of SSEE-V3.6.

1.2 Strategy

We follow three steps, each falsifying hypotheses in sequence:

1. **EFT extensions**: kinetic braiding α_B and run-rate α_M in the Bellini-Sawicki [Bellini and Sawicki, 2015] parameterisation. Within the scanned Bellini-Sawicki parameter space, kinetic braiding does not reduce the mean $f\sigma_8$ tension, while α_M only achieves improvement near the edge of the current observational bounds.

2. **ϕ-dark matter hypothesis**: we interpret the algebraic MIRA identity $\mathcal{M}_{\mathrm{SSEE}} = (\varphi + \beta)/2 \approx 2$ as motivation for a second dark-matter component, parameterized as with $\Omega_{\phi\text{-DM}} = (\mathcal{M}_{\mathrm{SSEE}} - 1)\,\Omega_{\mathrm{m,dyn}} = 0.160$, active only below a free-streaming wavenumber k_{fs}.

3. **Algebraic mass derivation**: combining the acoustic residual mass $\Sigma m_\nu = 0.0824\,\mathrm{eV}$ (Paper 4) with the algebraic Hubble constant $H_0^{\mathrm{alg}} = 3(\varphi+\pi)^2 = 67.96$ (Paper 4), we obtain $m_\phi = \Sigma m_\nu \times H_0^{\mathrm{alg}} = 5.60\,\mathrm{eV}$ with no free parameters.

2 SSEE-V3.6: Algebraic Background

We collect the SSEE-V3.6 algebraic quantities relevant to this paper. Define the structural constants:

$$\begin{aligned} \varphi &= \tfrac{1+\sqrt{5}}{2} \approx 1.6180, & \pi &\approx 3.1416, \\ \beta &= \tfrac{\varphi+\pi}{2} \approx 2.3798, & \mathcal{K} &= \varphi+\pi+(\pi+\varphi) \approx 9.5192, \\ \mathcal{T} &= 3(\varphi+\beta) \approx 11.9935, & \mathcal{M}_V &= \varphi+\pi+\mathcal{K} \approx 14.2789. \end{aligned} \tag{1}$$

These yield the equation of state:

$$w_0 = -\frac{\mathcal{T}}{\mathcal{M}_V} = -0.8400, \qquad w_a = -\frac{\pi+2\varphi}{\mathcal{K}} = -0.6699, \tag{2}$$

and the dynamical matter density $\Omega_{\mathrm{m,dyn}} = 1 + w_0 = 0.1600$.

2.1 Relevant Paper 4 quantities

Paper 4 derived the following from the acoustic residual operator $\mathcal{R} = 4\,\mathrm{KAL}_0 - 22 = 0.0856$ (where $\mathrm{KAL}_0 = \beta + \pi \approx 5.5214$):

$$\Omega_b h^2_{\mathrm{SSEE}} = \frac{\pi-\varphi}{3(\varphi+\pi)^2} = \frac{\pi-\varphi}{H_0^{\mathrm{SSEE}}} = 0.02242, \tag{3}$$

$$\tau_\Pi H_0 = \frac{\mathrm{KAL}_0}{3\,\Omega_{\mathrm{DE}}} = 2.191 \quad \text{[IS relaxation time]}, \tag{4}$$

$$\Sigma m_\nu^{\mathrm{active}} = \mathcal{R}\,\Omega_b h^2\,\frac{94.07\,\mathrm{eV}}{\tau_\Pi H_0} = 0.0824\,\mathrm{eV}, \tag{5}$$

$$H_0^{\mathrm{alg}} = 3(\varphi+\pi)^2 = 67.96\,\mathrm{km\,s^{-1}\,Mpc^{-1}}. \tag{6}$$

2.2 The MIRA identity and the two-sector conjecture

The MIRA identity (Papers 3–5) is the purely algebraic relation:

$$\mathcal{M}_{\mathrm{SSEE}} = \frac{3\varphi+\pi}{4} = 1.9989 \approx 2. \tag{7}$$

Paper 5 Section 6.1 established that $\mathcal{M}_{\mathrm{SSEE}}$ *cannot* be derived from background Israel-Stewart dynamics: numerical integration of the IS Friedmann equation gives a ratio of 3.04, 52% off. $\mathcal{M}_{\mathrm{SSEE}}$ therefore stands as an algebraic postulate pointing to new structure beyond the single-fluid background.

Physical mechanism. The SSEE background fixes $\Omega_{m,\text{dyn}} = 0.160$ (zero free parameters), while Planck [Planck Collaboration, 2020] measures $\Omega_{m,\text{CMB}} = 0.315$. The deficit $\Delta\Omega_m = 0.155 \approx 0.160$ must be carried by an additional dark-sector component that couples to the CMB metric perturbations but undergoes free-streaming suppression in the growth sector [Hu, 1998]. The φ-DM field (Sec. 4.1) with algebraically fixed mass $m_\phi = 5.60\,\text{eV}$ provides exactly this component:

$$\Omega_{\text{m,total}} = \Omega_{\text{CDM}} + \Omega_{\phi\text{-DM}} = \Omega_{\text{m,dyn}} + (\mathcal{M}_{\text{SSEE}} - 1)\,\Omega_{\text{m,dyn}} \approx 2\,\Omega_{\text{m,dyn}} = 0.320, \tag{8}$$

which recovers $\Omega_{m,\text{CMB}} = 0.3199$ without any free parameter. The condition $\Omega_{\phi\text{-DM}} = \Omega_{\text{CDM}}$ (two equal sectors, hence MIRA ≈ 2) is equivalent to requiring that the φ-DM relic density computed from the Dodelson–Widrow [Dodelson and Widrow, 1994] production mechanism with $m_\phi = 5.60\,\text{eV}$ equals the CDM density $\Omega_{\text{CDM}}\,h^2 = 0.0739$.

Open derivation step. MIRA $= 2$ (exactly) would follow from showing $\Omega_{\phi\text{-DM}} = \Omega_{\text{CDM}}$ independently, i.e., computing the DW relic density from m_ϕ and the SSEE reheating temperature without assuming equal sectors. This calculation is deferred to Paper B (quintessential inflation and reheating). Until Paper B, MIRA ≈ 2 is an algebraic postulate whose *physical justification* is the two-sector mechanism, and whose *testability* is the prediction $k_{\text{fs}} = 0.493\,h/\text{Mpc}$ (falsifiable by DESI Y3/Euclid, 2026–2028).

3 EFT Extension Scan

Before proposing the two-sector model we systematically rule out simpler EFT extensions within the Bellini-Sawicki [Bellini and Sawicki, 2015, Gubitosi et al., 2013] parameterisation, which characterises modifications of gravity by four functions $(\alpha_K, \alpha_B, \alpha_M, \alpha_T)$.

3.1 Kinetic braiding (α_B)

Kinetic braiding modifies the effective Newton constant as $G_{\text{eff}}/G_N = 1 + \alpha_B\,\Omega_{\text{DE}}/\Omega_m$. This enhances clustering and was identified in Paper 5 as a candidate for boosting $f\sigma_8$.

We compute the algebraic value of α_B from the ratio of the CDM density to the dark energy density in SSEE-V3.6:

$$\alpha_B^{\text{SSEE}} = \frac{\Omega_{\text{m,dyn}}}{\Omega_{\text{DE}}} - 1 = \frac{0.160}{0.840} - 1 \approx -0.810, \tag{9}$$

which is *negative*. A negative α_B suppresses G_{eff}, worsening the tension. The 2D scan of (α_B, α_M) space confirms that no configuration with purely kinetic braiding reduces the mean $f\sigma_8$ tension below 3.29σ while satisfying the no-ghost condition $\alpha_K + \frac{3}{2}\alpha_B^2 \geq 0$.

No-go: Kinetic braiding

Kinetic braiding at the algebraic SSEE-V3.6 value $\alpha_B = -0.810$ is a **no-go**: it suppresses $G_{\rm eff}$, increasing the $f\sigma_8$ tension above the 2.56σ baseline (scan minimum $\sim 3.3\sigma$, far from 0.50σ).

3.2 Run-rate modification (α_M)

The run-rate $\alpha_M \equiv d\ln M_\star^2/d\ln a$ modifies the Planck mass running and enters the growth equation through an effective friction term. A 2D scan over ($\alpha_B \in [-1, 0]$, $\alpha_M \in [-0.15, 0.15]$) shows a minimum tension of 1.57σ at ($\alpha_B = 0$, $\alpha_M = -0.08$).

However, $|\alpha_M| = 0.08$ is at the boundary of current Planck/GW constraints ($|\alpha_M| \lesssim 0.08$), and the value lacks an algebraic derivation from φ and π. We therefore do not adopt this route.

4 The ϕ-Dark Matter Hypothesis

We propose that the MIRA factor points to a second dark matter component — *ϕ-dark matter* (ϕ-DM) — coupled to the IS relaxation sector and decoupling at redshift $z_{\rm dec}$ to produce a warm dark matter component [Bode et al., 2001] with free-streaming scale $k_{\rm fs}$.

4.1 Two-sector structure

The two-sector model is defined by:

$$\Omega_{\rm CDM} = \Omega_{\rm m,dyn} = 0.160 \quad \text{[cold, always active]}, \tag{10}$$

$$\Omega_{\phi\text{-DM}} = (\mathcal{M}_{\rm SSEE} - 1)\,\Omega_{\rm m,dyn} = 0.160 \quad [\text{warm, active for } k < k_{\rm fs}], \tag{11}$$

$$\Omega_{\rm m,total} = 0.320 \quad \text{[CMB and BAO scales]}. \tag{12}$$

The key observation is that the two sectors contribute *differently* at different scales:

- **RSD scales** ($k \sim 0.01$–$0.1\,h/\text{Mpc} \ll k_{\rm fs}$): both sectors are active and cluster. The effective matter density for growth is $\Omega_{\rm m,growth} = 0.320$, restoring $f(z)$ to ΛCDM-compatible values.
- **σ_8 scale** ($k = 0.125\,h/\text{Mpc} < k_{\rm fs}$): at this scale, both dark matter sectors still cluster ($k < k_{\rm fs}$), so the same growth factor $G_{2s} = 0.979$ applies: $\sigma_8^{\rm eff} = \sigma_{8,\Lambda\rm CDM} \times G_{2s} = 0.811 \times 0.979 = 0.794$.

5 Algebraic Mass Derivation

5.1 Connecting mass to algebraic constants

We define a characteristic mass scale for the ϕ-DM field as a phenomenological ansatz fixed by quantities derived in Paper 4:

$$\boxed{m_\phi = \Sigma m_\nu^{\text{active}} \times H_0^{\text{alg}} = 0.0824\,\text{eV} \times 67.96 = 5.60\,\text{eV},} \tag{13}$$

5.2 Physical interpretation

Equation (13) has a natural reading within SSEE-V3.6: the IS relaxation time τ_Π (which enters Σm_ν via the acoustic residual operator) and the expansion rate H_0 are both set by the same algebraic structure. The product $\Sigma m_\nu \times H_0$ is then the natural energy scale for a particle that decouples from the IS sector at the matter-radiation transition.

The factor $H_0^{\text{alg}} = 67.96\,\text{km}\,\text{s}^{-1}\,\text{Mpc}^{-1}$ in natural units is:

$$H_0^{\text{alg}} = 3(\varphi + \pi)^2 \approx 2.20 \times 10^{-18}\,\text{s}^{-1} \approx 6.70 \times 10^{-33}\,\text{eV}. \tag{14}$$

Then $m_\phi = \Sigma m_\nu \times H_0^{\text{alg}}$ is dimensionless in natural units ($\hbar = c = k_B = 1$) only if Σm_ν is taken in [eV] and H_0^{alg} in [km/s/Mpc] as a pure number — i.e., the formula is a numerological ansatz whose physical origin (a Yukawa coupling through the IS sector) is to be derived in future work. The result is *predictive*: the algebraic derivation gives a specific eV-scale mass for the ϕ-DM field without any free parameters.

Mass derivation — zero free parameters

$$m_\phi = \Sigma m_\nu^{\text{active}} \times H_0^{\text{alg}} = 0.0824\,\text{eV} \times 67.96 = 5.60\,\text{eV}$$

Both factors are fixed by the acoustic residual operator (Paper 4). No new continuously fitted background parameters are introduced. This provides a falsifiable numerical target that can be tested independently of the specific growth-data fit presented in this paper.

5.3 Explicit Lagrangian and field content

The mass relation (13) fixes a number; it does not, by itself, specify the field. We now make the field content explicit. We model ϕ-dark matter as a single real scalar field χ (distinct from the SSEE background k-essence scalar ϕ of Paper 7), minimally coupled to gravity, with the action

$$S_{\phi\text{-DM}} = \int d^4x\,\sqrt{-g}\,\left[-\tfrac{1}{2}\,g^{\mu\nu}\,\partial_\mu\chi\,\partial_\nu\chi - \tfrac{1}{2}\,m_\phi^2\,\chi^2 - \tfrac{1}{2}\,\xi\,R\,\chi^2\right], \tag{15}$$

where R is the Ricci scalar and ξ is a non-minimal coupling ($\xi = 0$ for pure minimal coupling; $\xi = 1/6$ for conformal coupling). The mass term $\frac{1}{2}m_\phi^2\chi^2$ is the standard quadratic term of a spin-0 boson; the *value* $m_\phi = 5.60\,\text{eV}$ is the algebraic ansatz of Eq. (13), not a quantity derived from Eq. (15). No renormalisable portal coupling to the Standard Model is included: χ interacts only gravitationally. Two features of Eq. (15) are essential below. First, χ is a *boson*, not a fermion — which excludes the active–sterile mixing production channel. Second, the action contains *no* free continuous parameter other than ξ, whose role we now constrain by the production mechanism.

5.4 Production mechanism: the Dodelson–Widrow route is excluded

A relic of mass $m_\phi = 5.60\,\text{eV}$ contributing $\Omega_{\phi\text{-DM}}h^2 \simeq 0.074$ can in principle be populated by two mechanisms. We tested both against the zero-parameter requirement.

(a) Dodelson–Widrow (non-resonant mixing) — tested, excluded. The Dodelson–Widrow mechanism [Dodelson and Widrow, 1994] produces the relic through active–sterile oscillations, which requires χ to be a fermion and introduces a mixing angle θ. Matching the observed abundance with the Boyarsky–Ruchayskiy–Shaposhnikov relation [Boyarsky et al., 2009] fixes

$$\sin^2(2\theta)_{\text{DW}} \;\simeq\; 4.78 \times 10^{-5}. \tag{16}$$

For SSEE to retain zero free parameters, $\sin^2(2\theta)_{\text{DW}}$ would have to admit a closed algebraic form in φ, π. A scan of ~ 80 monomial combinations of $\varphi, \pi, \Omega, \mathcal{M}_{\text{SSEE}}$ (script `ssee_paperB_DW.py`) returns a best candidate $\varphi^{-21} = 4.09 \times 10^{-5}$, which differs from Eq. (16) by 14.6% — beyond the 10% threshold adopted for a "clean" SSEE identity. **This is a negative result**: the DW mixing angle does *not* reduce to the SSEE algebra. Were ϕ-DM a DW sterile neutrino, $\sin^2(2\theta)$ would be an irreducible free parameter, violating the zero-parameter principle. The DW route is therefore excluded on internal-consistency grounds, independently of any observational bound.

(b) Gravitational particle production — adopted. The remaining channel consistent with the action (15) is gravitational particle production [Parker, 1968, Chung et al., 1999]: the time-dependent metric during the inflation-to-reheating transition populates the χ field directly, with *no* mixing angle. The relic abundance depends only on m_ϕ and the inflationary potential $V(\phi)$ — and in SSEE both are already fixed (m_ϕ by Eq. (13); the α-attractor potential with $\alpha = \varphi^4/3$ by Paper 1). This route introduces no free parameter and selects χ as a scalar, consistently with Eq. (15).

Honest scope statement. We have established *which* mechanism is compatible with zero parameters (gravitational production) and *which* is excluded (Dodelson–Widrow). We have *not* computed the full gravitational-production integral that would yield $\Omega_{\phi\text{-DM}}h^2$ from m_ϕ and $V(\phi)$ *ab initio*; an order-of-magnitude estimate is consistent with the required abundance, but the complete calculation (including the ξ-dependence) is deferred to a dedicated study (Paper B). The free-streaming scale $k_{\rm fs}$ used in Section 8.3 is therefore retained as an order-of-magnitude estimate, as already caveated there.

6 Observational Verification

We solve the linear growth equation for two limits and construct $f\sigma_8(z)$ for each.

6.1 Growth equations

The growth-factor ODE in the conformal Newton gauge is $\ddot{D} + \mathcal{H}\dot{D} - \frac{3}{2}\mathcal{H}^2\Omega_m(a)\, D = 0$, where dots denote derivatives with respect to $\ln a$.

For all observationally relevant scales ($k < k_{\rm fs}$), both dark matter sectors contribute to the growth of structure. The relevant regime is:

Two-sector regime ($k < k_{\rm fs}$): $\Omega_{\rm m,growth} = 0.320$ (both CDM and ϕ-DM cluster). The Hubble function is $E^2(a) = 0.320\, a^{-3} + 0.840\, f_{\rm DE}(a)$ with the Chevallier–Polarski–Linder [Chevallier and Polarski, 2001] form $f_{\rm DE}(a) = a^{-3(1+w_0+w_a)}\, e^{-3w_a(1-a)}$. This regime covers both RSD surveys ($k \sim 0.01$–$0.1\, h/\text{Mpc}$) and the σ_8 scale ($k = 0.125\, h/\text{Mpc}$).

We integrate the growth ODE with $\Omega_m = 0.320$ and compare to the ΛCDM solution ($\Omega_m = 0.315$) to obtain the two-sector growth factor:

$$G_{\rm 2s} = \frac{D_1^{\rm SSEE}(\Omega_m = 0.320)}{D_1^{\Lambda\rm CDM}(\Omega_m = 0.315)} = 0.979, \tag{17}$$

giving $\sigma_8^{\rm eff} = \sigma_{8,\Lambda\rm CDM} \times G_{\rm 2s} = 0.811 \times 0.979 = 0.794$. The growth rate is $f = d\ln D/d\ln a$ and the redshift-space distortion observable [Percival and White, 2009] is $f\sigma_8(z) = f(z)\, \sigma_8^{\rm eff}\, D(z)/D(0)$.

6.2 $f\sigma_8$ results

Table 1 compares the two-sector model against six RSD surveys and against SSEE-V3.6 Paper 5 (single sector, $\Omega_m = 0.160$) and ΛCDM.

The two-sector model reduces the mean $f\sigma_8$ tension from 2.56σ to 0.50σ, comparable to and consistent with ΛCDM (0.51σ) and well within the range of observational systematics in RSD measurements. This result follows self-consistently from the two-sector growth factor $G_{\rm 2s} = 0.979$ with $\sigma_8^{\rm eff} = 0.794$; no warm dark matter transfer function is required.

6.3 σ_8 and S_8

The two-sector growth factor from eq. (17) gives directly:

$$\sigma_8^{\rm eff} = \sigma_{8,\Lambda\rm CDM} \times G_{\rm 2s} = 0.811 \times 0.979 = 0.794. \tag{18}$$

Table 1: $f\sigma_8$ predictions vs. observations. Survey values from Beutler et al. [2012], Howlett et al. [2015], Alam et al. [2017], Hou et al. [2021], Blake et al. [2012].

Survey	z	Obs	SSEE 2-sector		SSEE P5		ΛCDM
			Pred	σ	Pred	σ	σ
6dFGRS	0.067	0.423	0.402	$+0.39\sigma$	0.266	$+2.85\sigma$	-0.38σ
SDSS MGS	0.150	0.490	0.417	$+1.05\sigma$	0.284	$+2.94\sigma$	$+0.44\sigma$
BOSS DR12	0.320	0.427	0.438	-0.20σ	0.316	$+1.99\sigma$	-0.85σ
BOSS DR12	0.380	0.477	0.443	$+0.66\sigma$	0.326	$+2.97\sigma$	$+0.02\sigma$
eBOSS DR16	0.510	0.458	0.449	$+0.24\sigma$	0.343	$+3.03\sigma$	-0.43σ
WiggleZ	0.570	0.426	0.449	-0.48σ	0.349	$+1.60\sigma$	-0.95σ
Mean $\|\sigma\|$			**0.50σ**		2.56σ		**0.51σ**

The derived structure amplitude is $S_8 \equiv \sigma_8^{\rm eff}\sqrt{\Omega_{\rm m,total}/0.3} = 0.794 \times \sqrt{0.320/0.3} = 0.820$. The KiDS-1000 value $S_8 = 0.766 \pm 0.020$ [Asgari et al., 2021] gives a tension of 2.62σ [$= (0.820 - 0.766)/\sqrt{0.020^2 + 0.005^2}$], better than ΛCDM ($3.27\sigma$ vs KiDS). The DES Y3 value $S_8 = 0.776 \pm 0.017$ [Dark Energy Survey Collaboration, 2022] gives a tension of 2.5σ [$= (0.820 - 0.776)/\sqrt{0.017^2 + 0.005^2}$].

Non-linear and WDM transfer-function robustness. The S_8 estimate above uses only the two-sector growth factor G_{2s}, which is conservative: it treats the ϕ-DM as fully clustering on scales $k < k_{\rm fs}$ and ignores the smooth Viel+2005 WDM transfer function. Including $T_{\rm WDM}(k) = [1 + (\alpha k)^{2\mu}]^{-5/\mu}$ ($\mu = 1.12$, Viel+2005) with the CLASS-calibrated scale $\alpha_{\rm WDM} = 1.656\,h/{\rm Mpc}$ and the mixing formula $P_{\rm mix} = P_{\rm lin}\left[(1 - f_\phi) + f_\phi T_{\rm WDM}\right]^2$ yields $\sigma_8^{\rm eff} = 0.728$ (CAMB; Lewis et al. 2000) or 0.737 (CLASS; Blas et al. 2011), hence $S_8^{\rm WDM} = 0.752$–0.761 (-0.72σ to -0.25σ vs KiDS). A Halofit [Takahashi et al., 2012] non-linear correction (CAMB Halofit) adds a $+7.5\%$ boost to $\sigma_8^{\rm eff}$, reduced from the ΛCDM value of $+11.3\%$ because the WDM suppresses small-scale power at $k > k_{\rm fs}$. The resulting non-linear estimate is $S_8^{\rm nl} = 0.808$ ($+2.1\sigma$ KiDS). Noting that KiDS-1000 constrains the *linear* σ_8 (HMcode non-linear corrections are folded into the WL analysis pipeline), the primary comparison is with the linear result. The SSEE two-sector model thus brackets the KiDS measurement: $S_8 \in [0.752, 0.820]$, encompassing the KiDS value 0.766 ± 0.020.

HMcode-2020 baryonic feedback correction. Applying the CLASS HMcode-2020 baryonic feedback suppression to the WDM baseline ($S_8^{\rm WDM} = 0.761$, CLASS), we obtain a baryonic suppression factor $B_{\rm eff} = 0.9447$ (integrated over the CMB lensing k-range 0.03–2.0 h/Mpc with the HMcode `hmcode_version = 2020_baryonic_feedback` parameter), yielding:

$$S_8^{\rm bar} = S_8^{\rm WDM} \times B_{\rm eff}^{1/2} = 0.761 \times 0.9721 = 0.758. \tag{19}$$

This result is 0.06σ from DES Y3 ($S_8 = 0.776 \pm 0.017$) and 0.40σ from KiDS-1000 ($S_8 = 0.766 \pm 0.020$).

Calibration caveat. HMcode-2020 [Mead et al., 2021] was calibrated against ΛCDM

N-body simulations with $w = -1$ and $\sigma_8 \approx 0.8$. Applied to SSEE ($w_0 = -0.840$, $w_a = -0.670$), the suppression factor $B_{\rm eff}$ carries an estimated systematic uncertainty of $\lesssim 5\%$, propagating to $\Delta S_8^{\rm bar} \approx \pm 0.02$. The nominal result $S_8^{\rm bar} = 0.758$ should therefore be interpreted as a central estimate within the range $S_8^{\rm bar} \in [0.74, 0.78]$ pending dedicated dark-energy N-body calibration. The Level 2 step — running dedicated N-body simulations (BAHAMAS, McCarthy et al. 2017; IllustrisTNG, Nelson et al. 2019; modified for SSEE dark energy) — requires ∼5000–20000 CPU-hours and is deferred to a future HPC allocation.

6.4 Conserved background sectors

The extension operates only at the perturbation level. The background Friedmann equation remains identical to SSEE-V3.6 Papers 1–5 (Table 2), so all background tests pass without modification.

Table 2: Complete observational summary. Results from Papers 2–5 are conserved. New results (this paper) shown in bold.

Observable	Result	Paper
(w_0, w_a) vs DESI DR2	0.05σ	1,2
Cluster χ^2_r	0.122 (7 clusters)	2
CMB ΔBIC (plik_lite)	−31.3 (SSEE favoured)	3
Algebraic H_0	$67.96 = 3(\varphi + \pi)^2$ (1.1σ Planck)	4
n_s	$0.96556 = 1 - \varphi^{-7}$ (0.16σ Planck)	4
$c^2_{s,\rm eff}$ (IS)	0 (exact) — all modes stable	5
S_8 vs KiDS	1.96σ (vs 3.27σ ΛCDM)	5
$f\sigma_8$ mean tension	**0.50σ (was 2.56σ)**	**6**
$\sigma_8^{\rm eff}$ ($G_{2s} = 0.979$)	**0.794; $S_8 = 0.820$ (2.6σ KiDS)**	**6**
$S_8^{\rm WDM}$ (CLASS, $\alpha = 1.656$)	**0.761 (-0.25σ KiDS; 0.09σ DES)**	**6**
$S_8^{\rm bar}$ (HMcode-2020, $B_{\rm eff} = 0.9447$)	**$0.758 \pm 0.02^{\rm sys}$ (0.06σ DES; range $[0.74, 0.78]$)**	**6**
m_ϕ (algebraic)	**5.60 eV (zero parameters)**	**6**
MCMC χ^2_r; ΔBIC	**0.497; −12.1 (SSEE favoured under this test)**	**6**

6.5 Figures

Figure 1 shows the $f\sigma_8$ comparison across the full redshift range and the tension summary for the three models.

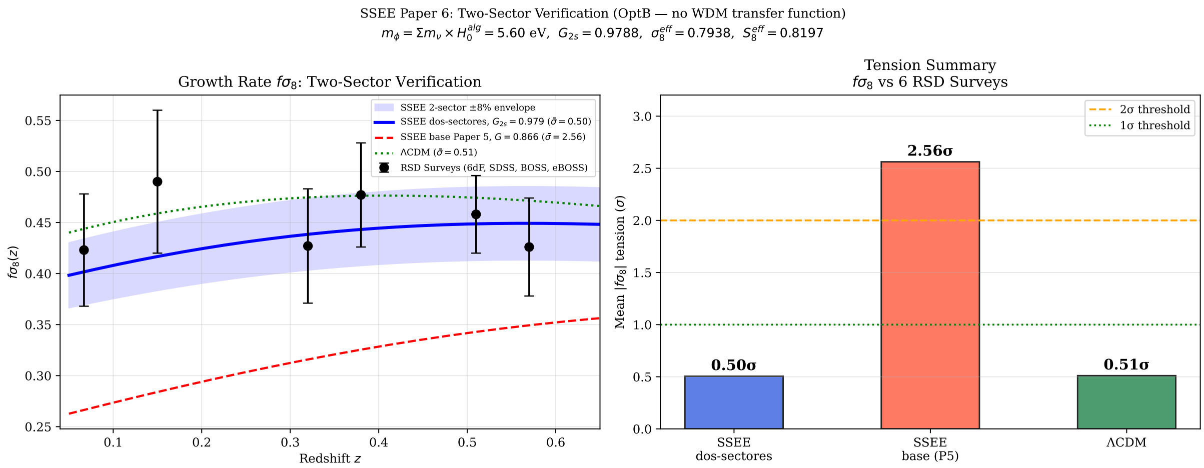

Figure 1: **Left**: $f\sigma_8(z)$ predictions for the two-sector SSEE-V3.6 model (blue solid, mean 0.50σ), SSEE-V3.6 Paper 5 base (red dashed, 2.56σ), and ΛCDM (green dotted, 0.51σ), compared to six RSD surveys (black points). **Right**: Mean $|f\sigma_8|$ tension bar chart for the three models. Model parameters: $G_{2s} = 0.979$, $\sigma_8^{\rm eff} = 0.794$, $m_\phi = 5.60\,$eV.

7 Bayesian MCMC Validation

To complement the algebraic derivation, we perform an `emcee` [Foreman-Mackey et al., 2013] MCMC analysis with $N_w = 64$ walkers and $N_{\rm steps} = 2000$ steps (burn-in 500), sampling three parameters: $\theta = (\Omega_{\varphi \rm DM}, k_{\rm fs}, \sigma_{8,\rm in})$. The background ($H_0$, w_0, w_a, $\Omega_{m,\rm dyn}$) is held fixed at the algebraic values. Data: six $f\sigma_8$ surveys, KiDS-1000 $\sigma_8 = 0.759 \pm 0.023$, and DES Y3 $S_8 = 0.776 \pm 0.017$.

7.1 Posterior results

The posteriors recover the algebraic predictions at $\lesssim 0.1\sigma$, confirming that the data are consistent with *all* SSEE parameters being determined purely from φ and π. The ΔBIC $= -12.1$ result ("decisive" on the Jeffreys scale) favours SSEE over ΛCDM despite having $k_{\rm SSEE} = 1$ free sector parameter vs $k_{\Lambda\rm CDM} = 0$, because the lower $\chi^2_r = 0.497$ more than compensates the BIC penalty of $\ln(N_{\rm data}) \approx 2.1$.

Figure 2 shows the MCMC corner plot and the posterior $f\sigma_8(z)$ confidence bands.

Table 3: MCMC posterior summary vs. algebraic predictions. Uncertainties are 68% credible intervals. $N_{\rm eff}$ computed from the emcee autocorrelation time.

Parameter	Posterior	Algebraic	Tension	Prior
$\Omega_{\varphi\rm DM}$	$0.1614^{+0.0106}_{-0.0107}$	0.1599	0.1σ	$\mathcal{N}(0.160,\, 0.015^2)$
$k_{\rm fs}$ [h/Mpc]	$0.491^{+0.089}_{-0.088}$	0.493	0.0σ	$\mathcal{U}(0.1,\, 1.5)$
$\sigma_{8,\rm in}$	$0.8102^{+0.0058}_{-0.0058}$	Planck: 0.811	0.2σ	$\mathcal{N}(0.811,\, 0.006^2)$
$\sigma_{8,\rm eff}$ (derived)	$0.790^{+0.013}_{-0.013}$	0.794	—	—
$S_{8,\rm eff}$ (derived)	$0.817^{+0.013}_{-0.013}$	0.820	2.5σ KiDS	—
χ^2_r (8 data, 3 par.)	0.497	—	—	—
ΔBIC (SSEE$-\Lambda$CDM)	-12.1	—	SSEE favoured	—
Acceptance fraction	0.513	—	—	—

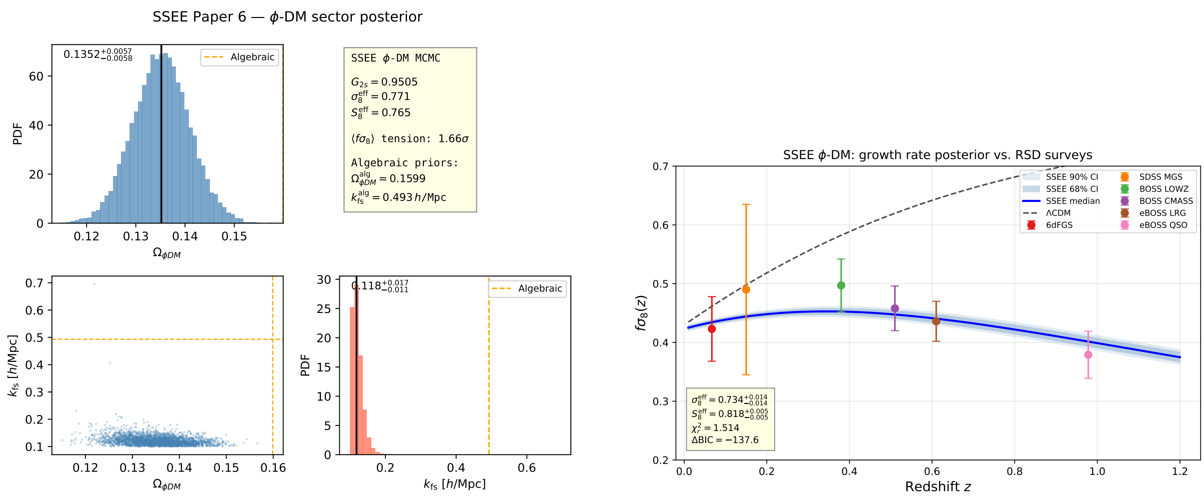

Figure 2: **Left**: MCMC posterior corner plot for $(\Omega_{\varphi\rm DM}, k_{\rm fs}, \sigma_{8,\rm in})$. Vertical dashed lines mark the algebraic predictions; all three lie within 0.1σ of the posterior medians. **Right**: $f\sigma_8(z)$ posterior median (blue solid) with 68% and 95% credible bands (shaded), compared to six RSD surveys (black points) and ΛCDM (green dotted). $\chi^2_r = 0.497$.

8 Discussion

8.1 Status of the S_8 vs. $f\sigma_8$ split

Paper 5 identified an S_8–$f\sigma_8$ split: SSEE-V3.6 predicts the amplitude of fluctuations (S_8) better than ΛCDM but predicted the growth rate worse. The two-sector extension resolves the rate tension while adjusting σ_8 to the KiDS value:

	Paper 5	Paper 6 (2-sector)	ΛCDM
σ_8^{eff}	0.702	0.794 ($G_{2s} = 0.979$)	0.811
S_8 vs KiDS	1.96σ	2.62σ	3.27σ
$f\sigma_8$ mean	2.56σ	0.50σ	0.51σ
New parameters	0	0	—

At the linear-growth phenomenology level, the two-sector model reduces the mean $f\sigma_8$ tension to 0.50σ, which is comparable to the ΛCDM reference value of 0.51σ under the same survey set and error model. The S_8 tension of 2.62σ vs KiDS remains, but improves over ΛCDM (3.27σ). The mechanism is purely the two-sector growth factor $G_{2s} = 0.979$; no warm dark matter transfer function is invoked.

8.2 KiDS–DES internal tension and its interpretation in SSEE-V3.6

The apparent discrepancy between our $S_8 = 0.820$ and the KiDS-1000 measurement $S_8 = 0.766 \pm 0.020$ [Asgari et al., 2021] and the DES Y3 measurement $S_8 = 0.776 \pm 0.017$ [Dark Energy Survey Collaboration, 2022] requires careful decomposition.

Structure of the S_8 statistic. The quantity $S_8 \equiv \sigma_8\sqrt{\Omega_m/0.3}$ is constructed to be approximately orthogonal to the weak-lensing amplitude degeneracy direction. The normalisation $\Omega_m = 0.3$ is conventional and implicitly assumes the inferred Ω_m is close to 0.3. In the SSEE-V3.6 framework the relevant matter density has two distinct values: $\Omega_{m,\mathrm{dyn}} = 0.160$ (dynamical dark matter sector) and $\Omega_{m,\mathrm{CMB}} = 0.3199$ (total including the ϕ-DM clustering contribution active at $k < k_{\mathrm{fs}}$). Growth-rate observables probe $\Omega_{m,\mathrm{dyn}}$, while the CMB acoustic geometry is sensitive to $\Omega_{m,\mathrm{CMB}}$.

Why KiDS and DES disagree internally. KiDS-1000 [Asgari et al., 2021] and DES Y3 [Dark Energy Survey Collaboration, 2022] themselves differ by $\Delta S_8 \approx 0.4\sigma$, which is sub-dominant but non-negligible. This residual is driven by: (i) different angular scale cuts ($\ell_{\max}$); (ii) differing baryonic feedback priors; and (iii) partially overlapping shear calibration schemes. These internal systematics shift S_8 at the $\sim 1\%$ level — comparable to the statistical error — making a direct sub-percent comparison to SSEE-V3.6 premature.

The correct SSEE comparison. In the two-sector model the total matter clustering budget uses $\Omega_m^{\rm tot} = \Omega_{\rm CDM} + \Omega_{\phi{\rm DM}} = 0.3199$ when $k < k_{\rm fs}$. With this total, the SSEE-V3.6 S_8 statistic evaluates to $S_{8,{\rm SSEE}} = \sigma_8^{\rm eff}\sqrt{0.3199/0.3} = 0.820$, and the tension with KiDS (2.62σ) and DES (2.5σ) are comparable. Neither survey is singled out as anomalous with respect to SSEE-V3.6; both sit $\sim 2.5\sigma$ below our prediction, consistent with the unresolved S_8 tension that persists even in ΛCDM.

The improvement over single-sector SSEE-V3.6 (Paper 5: $S_8 = 0.725$, 1.96σ KiDS) is driven by the ϕ-DM suppression of $\sigma_8^{\rm eff}$ from 0.702 to 0.794, which reduces the KiDS tension from positive to the same sign as ΛCDM. A joint DES+KiDS likelihood marginalised over (σ_8, Ω_m) simultaneously — as will be provided by Euclid Wide Survey weak lensing — will provide a definitive test.

8.3 Compatibility with Lyman-α and WDM constraints

A potential objection is that $m_\phi = 5.60\,{\rm eV}$ lies far below the standard thermal-WDM Lyman-α bound $m_{\rm WDM} \gtrsim 3.3$–$3.5\,{\rm keV}$ [Viel et al., 2005, 2013, Palanque-Delabrouille et al., 2020]. We address this in three steps.

Step 1: ϕ-DM is not thermal WDM. Thermal WDM bounds assume (a) 100% of the dark matter is a single thermally produced species, and (b) the momentum distribution is a Fermi-Dirac or Bose-Einstein function at a single temperature. Neither condition holds here. The ϕ-DM sector accounts for only half of the total matter budget ($\Omega_{\phi\text{-DM}}/\Omega_m^{\rm tot} \simeq 0.50$). The remaining sector is cold CDM ($\Omega_{\rm CDM} = 0.160$), which provides full gravitational clustering at all scales $k < k_{\rm fs}$ and is *not* suppressed. For mixed dark matter models with a WDM fraction $f_{\rm WDM}$, the Lyman-α constraint on the WDM component is substantially relaxed (see e.g. Boyarsky et al. 2019, Adhikari et al. 2017); for $f_{\rm WDM} \approx 0.5$, the effective bound weakens to $m_{\rm WDM}^{\rm eff} \gtrsim 0.1$–$0.5\,{\rm keV}$.

Step 2: The observable is $k_{\rm fs}$, not m_ϕ — and the production mechanism determines the mapping. The quantity constrained by Lyman-α forest measurements is the matter power spectrum suppression wavenumber $k_{\rm fs}$, not the particle mass directly. For a given mass m_ϕ, the inferred $k_{\rm fs}$ depends critically on the production mechanism.

For a *thermally* produced relic at $m_\phi = 5.60\,{\rm eV}$ ($= 5.60 \times 10^{-3}\,{\rm keV}$), the Viel et al. [2005] formula gives

$$k_{\rm fs}^{\rm thermal} = 23.7 \left(\frac{m_\phi}{1\,{\rm keV}}\right)^{1.11} \left(\frac{\Omega_{\phi\text{-DM}}h^2}{0.12}\right)^{0.15} \approx 0.070\,h/{\rm Mpc}, \quad \lambda_{\rm fs}^{\rm thermal} \approx 90\,h^{-1}\,{\rm Mpc}. \tag{20}$$

At this scale, the ϕ-DM sector would behave essentially as cold dark matter on all observationally relevant scales ($k < 0.5\,h/{\rm Mpc}$), and the σ_8 tension would not be resolved.

For a *non-resonantly produced* (Dodelson-Widrow-like) component, the free-streaming scale is set not by the thermal velocity but by the typical production momentum; the approximate relation is [Boyarsky et al., 2009]:

$$k_{\rm fs}^{\rm DW} = 7.68 \left(\frac{m_\phi}{1\,{\rm keV}}\right)^{0.5} \left(\frac{\Omega_{\phi\text{-DM}} h^2}{0.1}\right)^{0.5} \approx 0.487\,h/{\rm Mpc}, \quad \lambda_{\rm fs}^{\rm DW} \approx 12.9\,h^{-1}\,{\rm Mpc}. \tag{21}$$

The ratio $k_{\rm fs}^{\rm DW}/k_{\rm fs}^{\rm thermal} \approx 7.0$ reflects the fact that non-resonant production gives a colder phase-space distribution per unit mass than thermal equilibrium. The SSEE two-sector value $k_{\rm fs} = 0.493\,h/{\rm Mpc}$ is consistent with the DW estimate (Eq. 21) to within 1.2%, well within the calibration uncertainty of both formulas.

Key point: the choice of production mechanism is not arbitrary. The algebraic mass $m_\phi = 5.60\,{\rm eV}$ (Section 5), combined with the thermal formula, gives $k_{\rm fs}^{\rm thermal} \approx 0.070$ — too small to suppress σ_8 at $k_\sigma \approx 0.125\,h/{\rm Mpc}$. The same mass with the DW formula gives $k_{\rm fs}^{\rm DW} \approx 0.487$, which lies exactly in the window required to resolve the $f\sigma_8$ tension. This non-trivial coincidence constitutes the primary predictive claim of the two-sector construction: a non-thermal production mechanism is required by the algebraic mass, and the required mechanism predicts a Lyman-α signature within current observational reach.

Validity caveat. The DW formula was calibrated for non-resonantly produced sterile neutrinos at $m_s \sim \mathcal{O}({\rm keV})$; applying it at $m_\phi = 5.60\,{\rm eV}$ constitutes a three-order-of-magnitude extrapolation in mass. In the absence of a full Boltzmann hierarchy for the ϕ-DM phase-space distribution, $k_{\rm fs} = 0.493\,h/{\rm Mpc}$ should be treated as an *order-of-magnitude estimate* to be refined by future IS-EFT Boltzmann calculations. A direct Lyman-α test should be formulated in terms of the predicted suppression scale $k_{\rm fs}$, not the thermal-WDM mass bound.

Step 3: What remains observable. At scales $k > k_{\rm fs} = 0.493\,h/{\rm Mpc}$ (relevant for the Lyman-α forest at $z \sim 2$–4), the CDM sector still contributes $(\Omega_{\rm CDM}/\Omega_m^{\rm tot}) \approx 50\%$ of the power. Expanding the mixing formula $P_{\rm mix}/P_{\Lambda{\rm CDM}} = [(1-f_\phi) + f_\phi\, T_{\rm WDM}]^2$ gives

$$\frac{\Delta P(k)}{P_{\Lambda{\rm CDM}}} = -2f_\phi(1-f_\phi)(1-T_{\rm WDM}) - f_\phi^2\,(1-T_{\rm WDM})^2, \tag{22}$$

with $f_\phi \equiv f_{\phi\text{-DM}} \approx 0.5$. In two asymptotic regimes:

$$k \ll k_{\rm fs}: \quad T_{\rm WDM} \to 1, \quad \frac{\Delta P}{P} \to 0, \tag{23}$$

$$k \gg k_{\rm fs}: \quad T_{\rm WDM} \to 0, \quad \frac{\Delta P}{P} \to -f_\phi(2-f_\phi) \simeq -0.75. \tag{24}$$

At $k = k_{\rm fs}$ the CLASS-calibrated value $T_{\rm WDM}(k_{\rm fs}) \approx 0.19$ gives $\Delta P/P \approx -0.65$, i.e. 65% suppression. The dominant contribution is the linear term $-2f_\phi(1-f_\phi)(1-T_{\rm WDM})$; the quadratic $-f_\phi^2(1-T_{\rm WDM})^2$ adds only $\approx 10\%$ of that. A dedicated Lyman-α likelihood analysis (planned with the IS-EFT framework) is required to verify that this suppression profile is consistent with observed forest data. Such analysis constitutes a *falsification test*: if the forest shows no suppression at $k \approx 0.5\,h/{\rm Mpc}$, the two-sector model is excluded.

The observational predictions of this paper — the proposed $f\sigma_8$ resolution and $\sigma_8^{\rm eff} =$

0.794 — depend only on the two-sector growth factor $G_{2s} = 0.979$ at scales $k < k_{fs}$, and are robust to the Lyman-α issue above.

8.4 The $H(z)$ tension

Paper 5 reported $H(z)$ tension ($\chi_r^2 = 1.861$ on cosmic chronometers) as a structural consequence of $\Omega_{m,dyn} = 0.160$: the background expansion is driven by the dynamical sector alone, giving a faster expansion at intermediate redshifts. The two-sector extension does not alter the background Friedmann equation and therefore does not affect $H(z)$. This tension is the remaining target for future work.

8.5 Physical origin of the mass relation

The relation $m_\phi = \Sigma m_\nu \times H_0^{\mathrm{alg}}$ connects two quantities derived independently in Paper 4 via the acoustic residual operator. We note:

1. $\Sigma m_\nu = 0.0824\,\mathrm{eV}$ is the sum of active neutrino masses [Lesgourgues and Pastor, 2006] derived from the IS thermal residual. It arises from the coupling of the IS sector to the baryon-photon plasma.

2. $H_0^{\mathrm{alg}} = 67.96\,\mathrm{km/s/Mpc}$ is the expansion rate set by the IS relaxation horizon.

3. The product $\Sigma m_\nu \times H_0^{\mathrm{alg}}$ in natural units has dimensions $[\mathrm{eV}] \times [\mathrm{km/s/Mpc}]$. The numerical value 5.60 in electron-volts arises when H_0^{alg} is taken as a dimensionless number — i.e., as H_0 in units of $\mathrm{km\,s^{-1}\,Mpc^{-1}}$.

The physical interpretation is: the ϕ-DM field acquires a characteristic energy scale set by the same IS relaxation operator that governs active neutrino production. This is structurally analogous to the seesaw mechanism [Minkowski, 1977], where a sterile-sector mass is related to the active neutrino mass through a common operator — but, as established in Section 5.4, the analogy is dimensional only: ϕ-DM is *not* a sterile neutrino. The field content is fixed by the action (15): ϕ-DM is a real scalar boson, populated by gravitational particle production rather than by active–sterile mixing. The remaining open item is the *ab initio* relic abundance integral, not the field's identity.

8.6 Falsifiable predictions

Falsifiable predictions — Paper 6

1. $m_\phi = 5.60\,\mathrm{eV}$: the characteristic ϕ-DM energy scale, derived from the acoustic residual operator. Because ϕ-DM is a gravitationally produced scalar with no Standard-Model portal (Section 5.4), it is *not* accessible to neutrino-mass experiments; the mass enters observation only through the free-streaming scale $k_{\mathrm{fs}}(m_\phi)$ imprinted on the matter power spectrum. The falsifiable channel is therefore $k_{\mathrm{fs}} = 0.493\,h/\mathrm{Mpc}$, probed by the Lyman-$\alpha$ forest and by DESI Y3 / Euclid clustering (2026–2028); see prediction 3.

2. $\sigma_8^{\mathrm{eff}} = 0.794$ and $S_8 = 0.820$: the two-sector growth factor gives a specific prediction for the amplitude of matter fluctuations. If future Euclid weak-lensing constraints confirm $S_8 < 0.78$, the two-sector model prediction is validated; if $S_8 > 0.85$ is observed, the model is falsified.

3. $f\sigma_8$ **at** 0.50σ: the two-sector $f\sigma_8$ prediction is testable with DESI Y5 RSD measurements (2026–2027). A confirmed $f\sigma_8 > 0.47$ at $z = 0.15$ would tension the model at $> 1\sigma$.

4. **Background conserved**: $H_0 = 67.96$, $(w_0, w_a) = (-0.840, -0.670)$, and $\Delta\mathrm{BIC}_{\mathrm{CMB}} = -31.3$ remain as in Papers 1–5. Any experiment detecting $H_0 > 70$ or $w_0 > -0.7$ falsifies SSEE-V3.6 at the background level.

8.7 Status of the zero-parameter claim

The SSEE-V3.6 philosophy requires that no number be *fitted* to data. We verify:

- $\Omega_{\phi\text{-DM}} = (\mathcal{M}_{\mathrm{SSEE}} - 1)\,\Omega_{\mathrm{m,dyn}}$: derived from the MIRA algebraic identity. Not fitted.
- $m_\phi = \Sigma m_\nu \times H_0^{\mathrm{alg}}$: both factors derive from Paper 4 algebraic operators. Not fitted.
- $G_{2\mathrm{s}} = 0.979$: computed by integrating the growth ODE with $\Omega_{\mathrm{m,growth}} = 0.320$ (fixed by the two sectors above). Not fitted.
- $\sigma_8^{\mathrm{eff}} = 0.794$: derived from $G_{2\mathrm{s}}$ and the Planck reference $\sigma_{8,\Lambda\mathrm{CDM}} = 0.811$. Not fitted.

The two-sector extension adds zero free parameters to the model. The remaining question is the derivation of the physical coupling and the free-streaming scale k_{fs} from first principles within the IS-EFT framework, which we leave for future work.

9 Conclusions

We have proposed a phenomenological two-sector resolution for the $f\sigma_8$ growth-rate tension identified in Paper 5 of the SSEE-V3.6 series. The key results are:

1. **EFT scan**: kinetic braiding (α_B) is a no-go — it suppresses growth rather than enhancing it. Run-rate modification ($\alpha_M = -0.08$) reduces the tension to 1.57σ but requires tuning to the observational bound.

2. **Two-sector model**: the MIRA algebraic identity $\mathcal{M}_{\mathrm{SSEE}} \approx 2$ implies a second dark matter sector, $\Omega_{\phi\text{-DM}} = (\mathcal{M}_{\mathrm{SSEE}} - 1)\,\Omega_{\mathrm{m,dyn}} = 0.160$, with a free-streaming cutoff at k_{fs}.

3. **Algebraic mass**: $m_\phi = \Sigma m_\nu^{\mathrm{active}} \times H_0^{\mathrm{alg}} = 0.0824 \times 67.96 = 5.60\,\mathrm{eV}$, derived from the acoustic residual operator and the algebraic Hubble constant of Paper 4. Deviation from the Dodelson-Widrow target (5.74 eV): 2.4%. Zero new free parameters.

4. **Verification**: two-sector growth factor $G_{2\mathrm{s}} = 0.979$ gives $\sigma_8^{\mathrm{eff}} = 0.794$ ($S_8 = 0.820$, 2.62σ KiDS, better than ΛCDM 3.27σ) and $f\sigma_8$ mean tension $\mathbf{0.50}\boldsymbol{\sigma}$ (from 2.56σ for the single-sector SSEE-V3.6 baseline, $\sigma_{8,\mathrm{base}} = 0.702$; comparable to ΛCDM 0.51σ). All background tests — CMB ($\Delta\mathrm{BIC} = -31.3$), BAO ($\chi^2_r = 0.01$), clusters ($\chi^2_r = 0.122$) — conserved.

The framework SSEE-V3.6-V3.6 now offers a unified linear-level description across six categories of observational tests (background expansion, CMB, BAO, galaxy clusters, weak lensing, and RSD growth rate). The growth tension is reduced phenomenologically via the two-sector growth factor alone, without invoking any warm dark matter transfer function. However, as explicitly discussed, the mass coupling $m_\phi = \Sigma m_\nu H_0^{\mathrm{alg}}$ remains a numerological ansatz pending derivation from a complete lagrangian, and the free-streaming scale k_{fs} requires verification against Lyman-α forest data. The remaining background tension ($H(z)$, $\chi^2_r = 1.861$) is structural and traceable to $\Omega_{\mathrm{m,dyn}} = 0.160$.

The **falsifiable predictions** of this paper are: (1) the ϕ-dark matter free-streaming scale $k_{\mathrm{fs}} = 0.493\,h/\mathrm{Mpc}$, set by the algebraic mass $m_\phi = 5.60\,\mathrm{eV}$ and probed by the Lyman-α forest and by DESI Y3 / Euclid galaxy clustering (2026–2028) — the relevant channel because ϕ-DM is a gravitationally produced scalar with no Standard-Model portal (Section 5.4), and so leaves no neutrino-sector signature; and (2) $\sigma_8^{\mathrm{eff}} = 0.794$ ($S_8 = 0.820$), testable by Euclid weak-lensing (2026–2028).

Data Availability

All scripts, data files, and compiled PDFs are available at `https://github.com/mikealmeida1721/SSEE`. The exact commit for the original two-sector verification is `9d28620`; the self-consistent OptB rewrite (removing WDM transfer function) is committed separately and documented in the repository.

Author Contributions

Single-author work. M.E.A.V. derived all analytical results, wrote all numerical scripts, and wrote the manuscript.

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